LOAN APPROVAL PREDICTION USING MACHINE LEARNING

A PROJECT REPORT

Submitted by

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We feel immense pleasure to introduce “Loan Approval Prediction using Machine Learning” as our project.

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BONAFIDE CERTIFICATE

Certified that this project report LOAN APPROVAL PREDICTION USING MACHINE LEARNING submitted towards the fulfillment of BCA 5th semester department of computer application of Institute of Engineering and Management, was carried by **Parthib Ghosh(12023054006205), Md faizan hussain (12023054006175), Kunal Dutta(12023054006162), Koena Das (120230540061)** under the supervision of Mr. PARTHA KOLEY. The matter embodied in this project work is genuine and motivating for future work.

Wish him/her all success.

Signature of the Head of the Department

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ABSTRACT

Loan approval is a critical process in banking and financial institutions, involving a decision on whether to approve a customer's loan application. Traditionally, this process is based on manual review of financial and demographic factors. This project applies Machine Learning (ML) techniques to predict loan approval using customer data such as income, credit history, employment type, and loan amount.

The models used include Decision Tree, Logistic Regression, Random Forest, Support Vector Machine, K-Nearest Neighbor, and Naive Bayes. Data preprocessing, feature scaling, and evaluation metrics such as accuracy, precision, recall, and F1-score were used to assess model

performance. Among the algorithms, Gaussian Naive Bayes showed the highest accuracy, followed closely by KNN and Random Forest.

This project demonstrates the potential of ML to automate loan screening, reduce human error, and improve financial decision-making efficiency.

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Chapter 1: Introduction

Loan approval is a fundamental task for banks, requiring analysis of multiple parameters to determine a customer's eligibility. Delays or inaccuracies in decision-making can lead to financial loss or customer dissatisfaction.

With the rise of data availability and computing power, machine learning models can now be trained to predict loan approval based on historical data. Features like applicant income, employment status, credit history, loan amount, and co-applicant income are used to build predictive models that automate the approval process.

Keywords:

- Loan Prediction
- Credit Risk
- Machine Learning
- Financial Technology
- Classification Models

Headings:

- To use ML Techniques to predict Osteoporosis.
- To provide an easy-to-use User interface.
- To increase the accuracy of Osteoporosis prediction and analyze different conditions.

Chapter 2: Literature Review :

Loan prediction has seen growing interest with the rise of financial technology (FinTech). Traditional models used rule-based decision systems. Recent works apply ML models like Decision Trees, Logistic Regression, and Deep Learning to assess loan eligibility. Major

challenges include class imbalance (many rejected vs. few approved cases), data quality, and the interpretability of models. Studies have shown ML models significantly outperform manual systems when trained on quality datasets.

Materials :

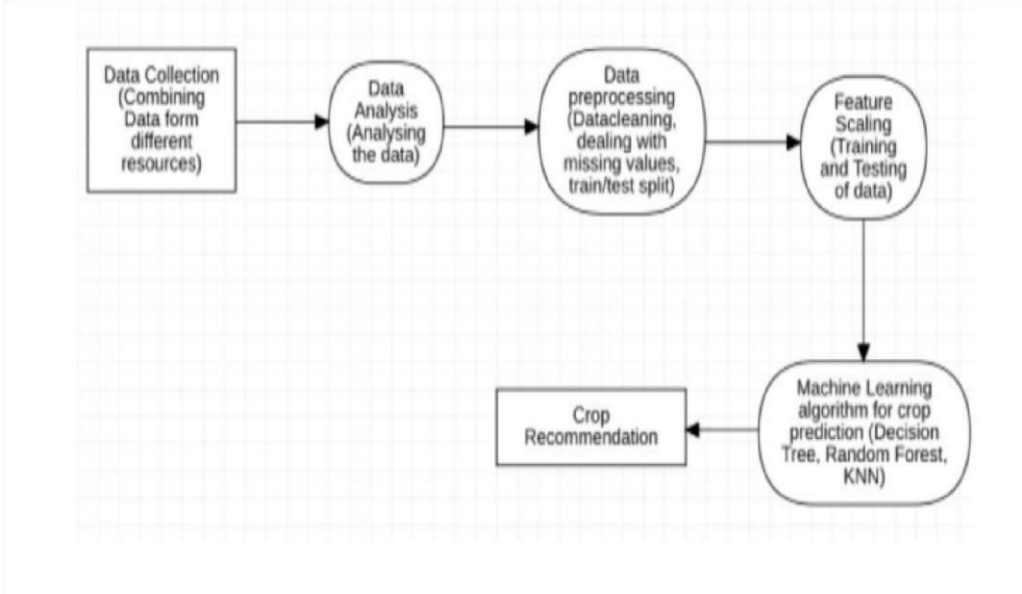
The dataset used for this project was sourced from publicly available platforms such as Kaggle, containing real-world examples of loan applications. Key features included:

- Applicant Income
- Loan Amount
- Credit History
- Marital Status
- Education Level

Chapter 3: Methodology

Proposed System

The Proposed System will predict the occurrence of Loan Approval on the basis of parameters like Gender, Age, Credit History, Loan Amount, Education , Marital Status.



In the above flow chart as mentioned we first collected the data from various places such as government data, kaggle and some private data, after collecting the data we analyse it that how every parameter matters in the prediction of loan approval and on the third step we preprocess our data i.e cleaning, data missing problem .Next we move on to the fourth section i.e the feature scaling in which we separate our main dataset into two datasets in which now of them is to train our dataset and other is for testing so that we can predict our accuracy after that we move on to the machine learning algorithm for loan approval prediction where we use many different algorithms for predicting loan approval and when every step is finished we move on out final phase i.e predicting loan approval.

Proposed System follows:

Collection of Data: The collection of data is extremely important in machine learning. In order for a machine learning model to be effective, it needs to be trained on a large and diverse dataset. If the data is of poor quality or is not representative of the problem the model is trying to solve, the model may not be

able to make accurate predictions or take appropriate actions . So We collect the data which contains parameters such as Age, Gender, Education, Credit History, Loan Amount from various places such as government websites, Kaggle, and some private data and place them into one dataset by understanding every dataset and put them together when every data set is ready to have the same exact column name with no duplicated data.

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	\
1	LP001003	Male	Yes	1	Graduate	No	
2	LP001005	Male	Yes	0	Graduate	Yes	
3	LP001006	Male	Yes	0	Not Graduate	No	
4	LP001008	Male	No	0	Graduate	No	
5	LP001011	Male	Yes	2	Graduate	Yes	
..	...	...	...	...	...	...	
609	LP002978	Female	No	0	Graduate	No	
610	LP002979	Male	Yes	3+	Graduate	No	
611	LP002983	Male	Yes	1	Graduate	No	
612	LP002984	Male	Yes	2	Graduate	No	
613	LP002990	Female	No	0	Graduate	Yes	

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	\
1	4583	1508.0	128.0	360.0	
2	3000	0.0	66.0	360.0	
3	2583	2358.0	120.0	360.0	
4	6000	0.0	141.0	360.0	
5	5417	4196.0	267.0	360.0	
..	...	...	...	...	
609	2900	0.0	71.0	360.0	
610	4106	0.0	40.0	180.0	
611	8072	240.0	253.0	360.0	
612	7583	0.0	187.0	360.0	
613	4583	0.0	133.0	360.0	

Analyzing of Data : In order to find patterns and trends that may be utilised to train the model, data analysis is a crucial phase in the crop prediction project. We can choose the most pertinent and practical elements to incorporate into the model by carefully examining the data, and we can also see any potential issues or biases that can affect the model's performance. This makes it possible for us to create a model that is more precise and efficient and that can anticipate loan approval prediction with greater reliability. We analyze the data by using many python libraries like Matplotlib, Seaborn, NumPy and pandas by plotting different graphs with many parameters to analyze the correlation between them and to know more about the importance of every parameter.

Data Preprocessing: In order to find patterns and trends that may be utilized to train the model, data analysis is a crucial phase in the loan approval prediction project. We can choose the most pertinent and practical elements to incorporate

into the model by carefully examining the data, and we can also see any potential issues or biases that can affect the model's performance. This makes it possible for us to create a model that is more precise and efficient and that can anticipate loan approval prediction with greater reliability. It plays a key role in developing a model that is accurate, reliable, and efficient. By carefully preprocessing the data, we can improve the performance of the model and ensure that it is able to make accurate predictions about loan approval prediction.

Feature Scaling: We can change the data so that each feature is on a comparable scale by using feature scaling. Several techniques, including normalization standardization, and min-max scaling, can be used to accomplish this. We can increase the data's suitability for machine learning techniques and the model's accuracy and dependability by scaling the features. After Data Preprocessing, we have to scale the dataset into two parts : training and testing data which help to find out the accuracy of the model.

Model Selection: We can choose the best algorithms to utilize for the crop prediction project by conducting a thorough examination of the data and the issue we are attempting to solve. This may entail contrasting various algorithms according to how well they work on the data, how sophisticated they are, and how well they generalize to new data. We can make sure that the model is able to estimate crop growth accurately by choosing the most suitable algorithm. In Model selection we select KNN, Decision Tree and Random Forest, Logistic Regression, Naive Bayes, Linear SVM . After selecting 3 different models we implement every single one and at the end we take the Decision Tree as it predicts more accurately than the other two algorithms.

Loan Approval Prediction : The system will suggest the best crop for cultivation based on the amount of anticipated rainfall, the composition of the soil, and weather parameters. This approach also displays the amount of seed needed to cultivate a recommended crop in parts per million. weather parameters. This approach also displays the amount of seed needed to cultivate a recommended crop in parts per million.

Confusion Matrix and Classification Report:

Confusion Matrix and Classification Report are the methods imported from the metrics module in the scikit learn library that are calculated using the actual labels of test datasets and predicted values. Confusion Matrix gives the matrix of frequency of true negatives, false negatives, true positives and false positives. Classification Report is a metric used for evaluating the performance of a classification algorithm's predictions.
It gives three things: Precision, Recall and f1-score of the model.

Precision refers to a classifier's ability to identify the number of positive predictions which are relatively correct. It is calculated as the ratio of true positives to the sum of true and false positives for each class.

$$Precision = \frac{TP}{TP + FP}$$

Where Precision-Positive Prediction Accuracy; TP-True Positive; FP-False Positive.
Recall is the capability of a classifier to discover all positive cases from the confusion matrix. It is calculated as the ratio of true positives to the sum of true positives and false negatives for each class.

$$Recall = \frac{TP}{TP + FN}$$

Where Recall- The percentage of positives that were correctly identified;
FNFalse Negative.
F1 score is a weighted harmonic mean of precision and recall, with 0.0 being the worst and 1.0 being the best. Since precision and recall are used in the computation, F1 scores are often lower than accuracy measurements.

Where P-Precision; R-Recall

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

Accuracy: The number of correct predictions divided by the total number of predictions accuracy. The accuracy of the model is calculated using the accuracy_score method of scikit learn module.

Where TP-True Positive; FP-False Positive; TN : True Negative; FN : False Negative

$$\text{F1 score} = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})}$$

Chapter 4: Proposed Work Proposed Work:

K-Nearest Neighbour Classifier :

The K-Nearest Neighbour algorithm is based on the supervised learning technique and is a simple machine learning algorithm. The K-NN technique saves all possible and classifies the incoming data depending on how similar they are to the actual data. This means that the K-NN technique can swiftly classify new instances into a precisely defined category. The KNN technique can be used in both regression and classification problems but it is most likely to be used in classification.

KNN technique has two properties. First, the model is based on the dataset or, it is not required to identify parameters for the distribution. Hence it is referred to as non-parametric. Second, there is no learning taking place; instead, it just stores the training data. The classification of the dataset happens during the testing phase, due to which the testing phase becomes time-consuming and takes a lot of memory. This property is known as lazy-learner.

Step 1: K, i.e., the number of neighbours is selected. The primary deciding factor is the number of neighbours.

Step 2: Using distance measures, determine the distance between two points like Euclidean distance.

$$Euclidean\ distance = d(b,a) = \sqrt{\sum_{i=1}^{n}(b_i - a_i)^2}$$

Step 3: K nearest neighbours are taken into account according to the calculated Euclidean distance.

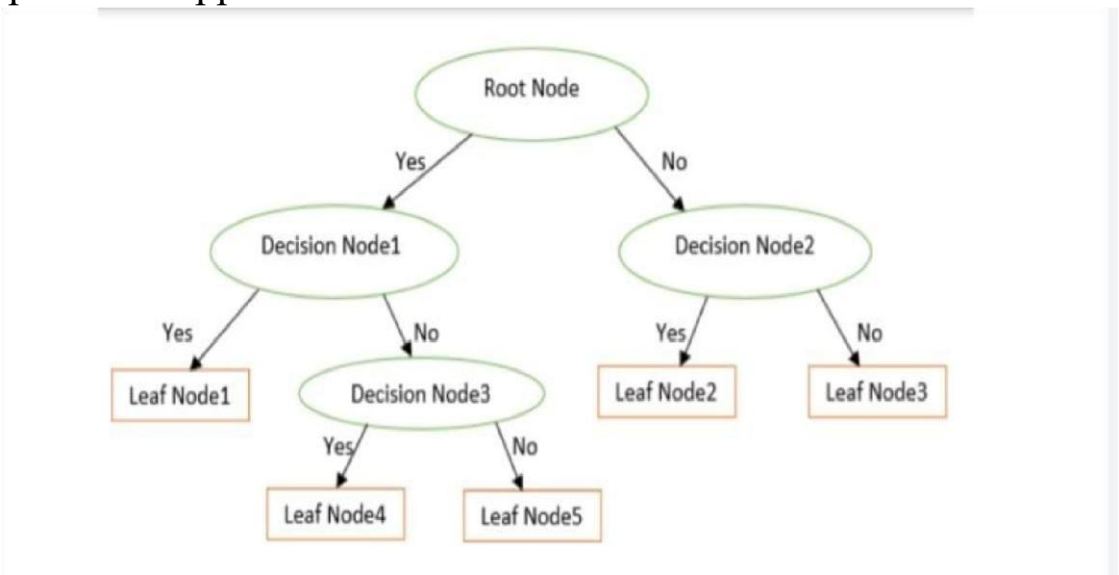
Step 4: Figure the number of data points in each class surrounded by these
Step 5: The class with the highest number of neighbours is assigned to the new data points.
Step 6: The label is voted on, and the model is ready.

K-Nearest Neighbors (KNN) Model:

- **Accuracy**: 86.39%
- **Precision**: The KNN model performs well in predicting both classes. Class 1 (eligible for loan) has a high precision (0.97), meaning the model correctly identifies eligible peoples most of the time.
- **Recall**: Class 0 (not eligible) has a very high recall (0.97), indicating that the model rarely misses identifying eligible individuals.
- **F1-Score**: The F1-score is balanced, with a slightly higher score for class 0, showing a good balance between precision and recall.

Decision Tree Classifier :

Decision Tree is a supervised learning technique used where each path is a set of decision instances from the training set. The non attributes. The decision is made by comparing the instance with the split and jumping to the next node. This splitting continues, generating sub nodes which determine class labels for that instance. It divides recursive partitioning. With high accuracy, decision trees are capable of handling high data. It's a flowchart diagram-style representation that closely parallels human result, decision trees are simple to explain and apprehend.



Step 1: Starting with the root node of the tree, which consists of the
Step 2: The most appropriate attribute is obtained from the dataset by applying the Attribute Selection Measure (ASM).

Step 3: The S is divided into subdivisions that attributes.

Step 4: The node is formed in the decision tree with the most appropriate attribute.

Step 5: The tree formation is setup by iteratively repeating this method for each child until one of the following requirements is met:

- The tuples are entirely correlated with the same attribute value.
- There are no further attributes accessible.
- There aren't any more instances.

Decision Tree (DT) Model:

- **Accuracy**: 82.14%
- **Precision**: Precision for class 0 is 0.83, and class 1 is 0.81, indicating that the model is fairly accurate but not as robust as others in avoiding false positives.
- **Recall**: Class 1 has a recall of 0.85, suggesting that the model is good at identifying eligible persons, but class 0 (not eligible) has a recall of 0.79, meaning it misses some not eligible individuals.
- **F1-Score**: The F1-scores are balanced for both classes, reflecting the model's ability to handle both classes with similar accuracy.

Steps to Split: The dataset used in the project has with the numerical values in the following ways:

Step 1: Sorting all the values.

Step 2: It will consider a threshold value from

Step 3: Feature value will split into two parts such that threshold value, and the right node contains feature values greater than

Step 4: The next feature value will be considered as a threshold value and again create the same split.

Step 5: Entropy/Gini and Information Gain are split with better information gain is considered.

$$I(Attribute) = \frac{\sum p_i + n_i}{p + n}$$

Where pi denotes the number of yes values; n attribute; p and n are the numbers of yeses and noes of the entire sample, respectively.

$$Information\ Gain = Entropy(S) - I(Attribute)$$

Where Entropy(S) denotes entropy of sample S; I(Attribute) denotes Average Information of the particular attribute.

Step 6: Repeat from Step 2 to Step 5. In this way it will get branches for the decision tree .

Entropy and Gini Index : The criteria for measuring Information Gain are the Gini index and Entropy. Information gain is a measurement of how much the reduction in entropy. Entropy and Gini Index are the metrics that measure the impurity of the nodes. A node is considered as impure if it has multiple classes, else Entropy is a metric that gives the degree of impurity in a specified attribute.

The following formula can be used to compute entropy:

Entropy : is a metric that gives the degree of impurity in a specified attribute.

The following formula can be used to compute entropy:

$$Entropy(S) = -P(yes)\ log_2\ P(yes) - P(no)\ log2\ P(no)$$

$$Entropy(S) = \sum_{i=1}^{n} -p_i log_2 p_i$$

Where S denotes a complete sample, P(yes) denotes the probability of yes , P(no) denotes the probability of no .

Gini Index: Gini is estimated by deducting the sum of squared probabilities of each class from one. The lower Gini Index value is preferred rather than a higher value. Scikit is the default value and supports “Gini” criteria for Gini

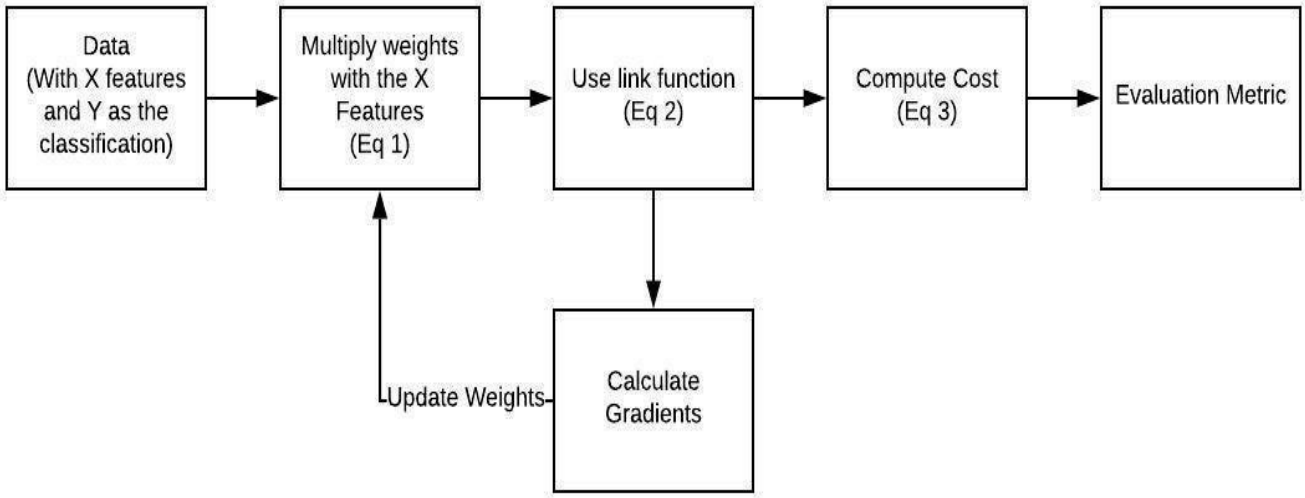
$$Gini\ Index = 1 - \sum_{i=1} (P_i^2)$$

Index.

Where Pi denotes the probability of a tuple , say R belonging to class Ci .

Logistic Regression:

Logistic Regression is a classification algorithm used to predict the probability of a binary outcome. It is based on the logistic function, which transforms linear output into a probability between 0 and 1.



Steps in Logistic Regression:

Step 1: Prepare the dataset by splitting it into training and testing sets. Handle missing values and encode categorical features.

Step 2: Apply the logistic function to the linear combination of input features, generating probabilities for class 1 (positive class).

$$P(y=1|X) = \frac{1}{1+e^{-(b_0+b_1X_1+b_2X_2+...+b_nX_n)}}$$

Step 3: Train the model using a method like Maximum Likelihood Estimation (MLE) to find the optimal coefficients (weights) that maximize the likelihood of the observed data.

Step 4: For each new instance, calculate the probability using the logistic function and compare it to a threshold (typically 0.5) to classify the instance into class 0 or class 1.

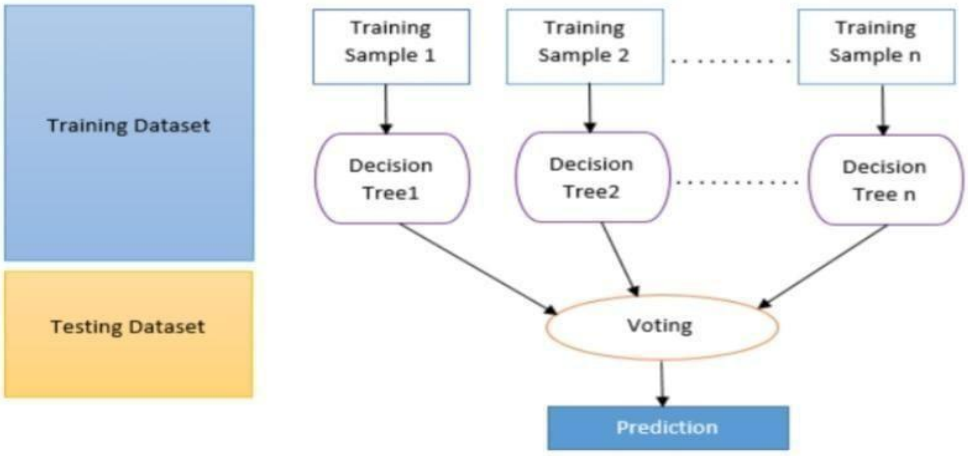
Step 5: Evaluate the model's performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to measure how well it predicts the outcomes.

Logistic Regression (LogReg) Model:

- **Accuracy**: 83.42%
- **Precision**: Class 1(eligible) has a slightly higher precision (0.88), meaning the model is effective in identifying.
- **Recall**: The recall for class 1 is 0.78, which is decent but not as high as for class 0 (0.89), indicating the model misses some loan approval cases.
- **F1-Score**: The F1-scores for both classes are similar, reflecting a balanced performance in predicting both classes.

Random Forest Classifier:

The Random Forest method consists of multiple decision tree classifiers to enhance the model's performance. Here ensemble learning is a supervised learning algorithm. Decision trees are created at random using the instances from the training set. Each of the decision trees the model is decided by majority voting. One of the reasons for its popularity as a machine learning approach is that it can handle the issue of overfitting trees.



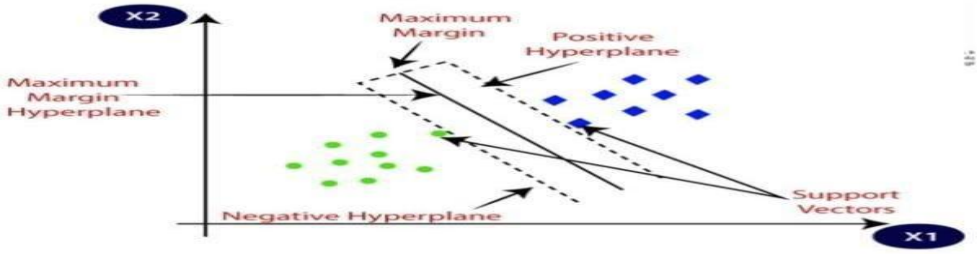
Step 1: K instances are chosen at random from the given training Step 2: Decision trees are created for the chosen instances.
Step 3: The N is selected for the number of estimators to be created.
Step 4: Here Step 1 & Step 2 is repeated.
Step 5: For the new instance, the predictions of each estimator is determined, and the category with the Hughes votes is assigned .

. **Random Forest (RF) Model:**

- **Accuracy**: 84.52%
- **Precision**: Class 1 has a very high precision of 0.97, meaning the model is very effective at correctly identifying eligible peoples .
- **Recall**: Class 0 has a very high recall (0.97), showing that it is excellent at identifying individuals eligible.
- **F1-Score**: Both precision and recall are balanced, providing an overall wellperforming F1-score for both classes

SVM :

SVM is used for Classification as well as Regression problems .The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane. SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine.



Here is this diagram of two different categories that are classified using a decision boundary or hyperplane .

Type of SVM :

1. Linear
2. Non -Linear

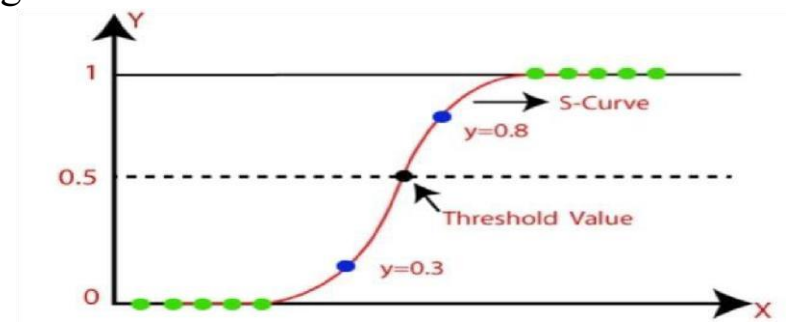
Here we will use a linear SVM for prediction .

Logistic Regression :

Logistic Regression is used for predicting the categorical dependent variable using a given set of independent variables.The outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, True or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).

Logistic Regression Equation :

The Logistic regression equation can be obtained from the Linear Regression equation. The mathematical steps to get Logistic Regression equations are given below:



We know the equation of the straight line can be written as:

$$log\left[\frac{y}{1-y}\right] = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_nx_n$$

In Logistic Regression y can be between 0 and 1 only, so for this let's divide the above equation by (1-y):

$\frac{y}{1-y}$; 0 for y= 0, and infinity for y=1

But we need range between -[infinity] to +[infinity], then take logarithm of the equation it will become:

$$log\left[\frac{y}{1-y}\right] = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_nx_n$$

The above equation is the final equation for Logistic Regression.
Steps in Logistic Regression:

- Data Preprocessing step
- Fitting Logistic Regression to the Training set
- Predicting the test result
- Test accuracy of the result(Creation of Confusion matrix) ● Visualizing the test set result.

Support Vector Classifier (SVC) Model:

- **Accuracy**: 83.16%
- **Precision**: Class 1 (eligible) has a precision of 0.89, and class 0 has a precision of 0.79, indicating that the model is fairly good at correctly identifying both classes.
- **Recall**: Class 1 (eligible) has a recall of 0.77, which means the model misses some eligible cases. Class 0 has a higher recall of 0.90, showing strong performance in identifying not eligible individuals.
- **F1-Score**: The F1-scores are balanced, with a slight advantage for class 0.

Naive Bayes :

○ The Naive Bayes algorithm is a supervised learning algorithm, which is based on Bayes theorem and used for solving classification problems.
○ It is mainly used in text classification that includes a high-dimensional training dataset.
○ It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.
○ Some popular examples of Naïve Bayes Algorithm are spam filtration, Sentimental analysis, and classifying articles.

Bayes' Theorem:

○ Bayes' theorem is also known as Bayes' Rule or Bayes' law, which is used to determine the probability of a hypothesis with prior knowledge. It depends on the conditional probability.

○ The formula for Bayes' theorem is given as:

$$P(A|B)=\frac{P(B|A)P(A)}{P(B)}$$

Where,

P(A|B) is Posterior probability: Probability of hypothesis A on the observed event B.

P(B|A) is Likelihood probability: Probability of the evidence given that the probability of a hypothesis is true.

P(A) is Prior Probability: Probability of hypothesis before observing the evidence.

P(B) is Marginal Probability: Probability of Evidence.

Working of Naïve Bayes' Classifier :

Working of Naïve Bayes' Classifier can be understood with the help of the below example:

Suppose we have a dataset of weather conditions and corresponding target variable "Play". So using this dataset we need to decide whether we should play or not on a particular day according to the weather conditions. So to solve this problem, we need to follow the below steps:

1. Convert the given dataset into frequency tables.
2. Generate Likelihood table by finding the probabilities of given features.
3. Now, use Bayes theorem to calculate the posterior probability.

Problem: If the weather is sunny, then the Player should play or not? Solution:

To solve this, first consider the below dataset:

	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No

5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Frequency table for the Weather Conditions:

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	5

Likelihood table weather condition:

Weather	No	Yes	
Overcast	0	5	5/14= 0.35
Rainy	2	2	4/14=0.29
Sunny	2	3	5/14=0.35
All	4/14=0.29	10/14=0.71	

Applying Bayes Theorem :

P(Yes|Sunny)= P(Sunny|Yes)*P(Yes)/P(Sunny)
P(Sunny|Yes)= 3/10= 0.3
P(Sunny)= 0.35 P(Yes)=0.71

Naive Bayes (NB) Model:

- **Accuracy**: 86.99%
- **Precision**: The Gaussian Naive Bayes model shows excellent precision for class 1 (eligible) at 0.97, meaning it rarely misclassifies not eligible individuals getting loan .
- **Recall**: Class 0 (not eligible) has a high recall of 0.97, showing that it effectively identifies individuals who are not eligible .
- **F1-Score**: Both classes have strong F1-scores, with class 0 having slightly better performance than class 1.

Chapter 5: Result And Discussion

Loan Approval Prediction is a condition characterized by credit history. Prediction of loan approval can significantly improve someones status by

providing money. In this project, machine learning (ML) algorithms such as K-Nearest Neighbors (KNN), Logistic Regression, Support Vector Machine (SVM), Decision Tree, and Random Forest are used to predict the likelihood of osteoporosis based on various input features (e.g., age, gender, credit history, etc.).

The goal is to evaluate and compare the performance of each of these algorithms in terms of prediction accuracy.

K-Nearest Neighbors (KNN): KNN is a non-parametric method that classifies data points based on the majority vote of their neighbors. The number of neighbors (K) is a hyperparameter.

Logistic Regression: Logistic Regression is a linear classifier that models the probability of the binary outcome (e.g., osteoporosis or not). It uses the logistic function (sigmoid) to predict probabilities.

Support Vector Machine (SVM): SVM aims to find the hyperplane that best separates the classes by maximizing the margin between the classes. For nonlinearly separable data, the kernel trick is used to map the data to a higherdimensional space.

Decision Tree: A decision tree splits the data into subsets based on the feature that provides the best split according to a criterion like Gini impurity or entropy. It creates a tree-like structure with branches and leaves.

Random Forest: Random Forest is an ensemble learning method that builds multiple decision trees using random subsets of the data and features. It aggregates the predictions of all the trees to make a final decision.

Here in this project the accuracy of K-Nearest Neighbors (KNN) is higher among other algorithms.

Table of Loan Approval:

Loan Approval Prediction – Loan Approval prediction by ‘0’ and ‘1’ Dataset - The source of the data used for the study.

Feature selection - The features or variables used to train the machine learning algorithm.

Machine learning algorithm - The name of the algorithm used for loan approval prediction.

Model accuracy - The accuracy of the model in predicting loan approval (usually in percentage).

Evaluation metric - The metric used to evaluate the performance of the model (e.g., RMSE, MAE, R^2).

Experiment setup - The parameters used for the experiment, such as training and testingsplit, hyperparameters tuning, and cross-validation.

Accuracy:

The accuracy of loan approval prediction using machine learning algorithms depends on various factors such as the quality and quantity of data used, the choice of machine learning algorithm, and the specific loan approval conditions and other involving factors. Generally, the accuracy of the prediction is evaluated using evaluation metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).

Additionally, techniques such as cross-validation help ensure the model's generalizability, preventing overfitting. Factors like feature selection, data quality, and handling missing or imbalanced data also significantly impact the model's performance. In clinical settings, it is essential for the model to not only be accurate but also interpretable and reliable, ensuring that it provides actionable insights for healthcare providers. Therefore, an accurate loan approval prediction model must be carefully assessed through multiple metrics and validated in clinical contexts to ensure its efficacy and safety.

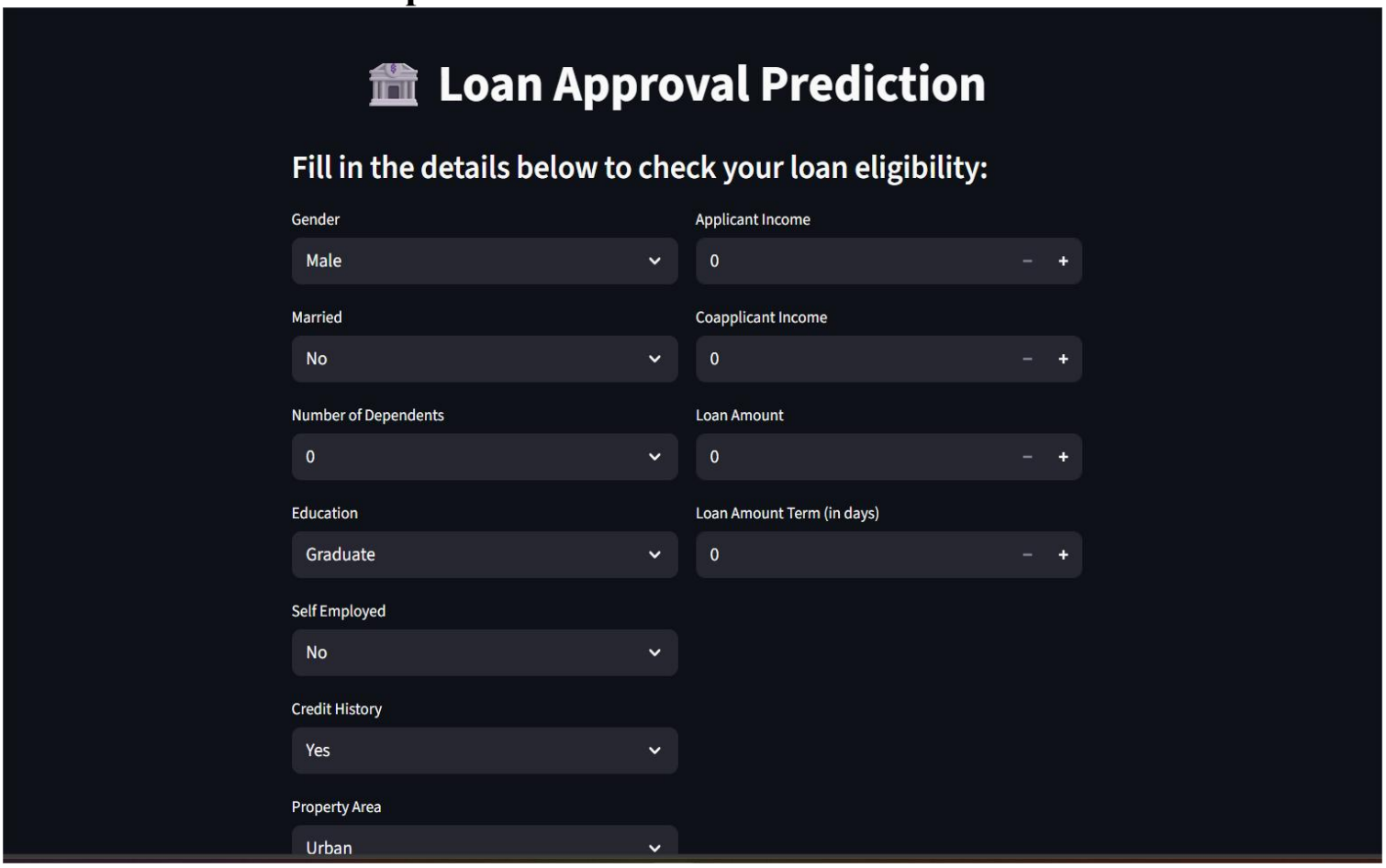
1. K-Nearest Neighbors (KNN)**:** 86.39%
2. Logistic Regression (LogReg)**:** 83.42%
3. Gaussian Naive Bayes (NB)**:** 86.99%
4. Decision Tree (DT)**:** 82.14%
5. Random Forest (RF)**:** 84.52%
6. Support Vector Classifier (SVC)**:** 83.16%

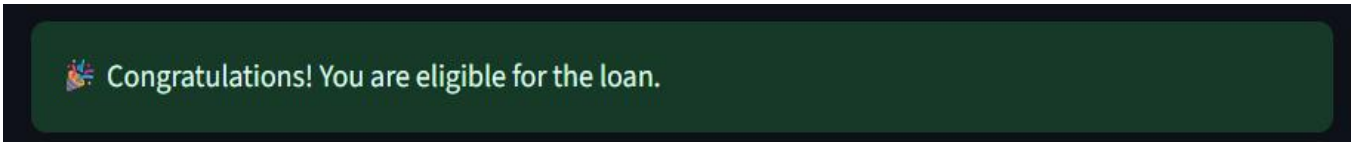
Summary:

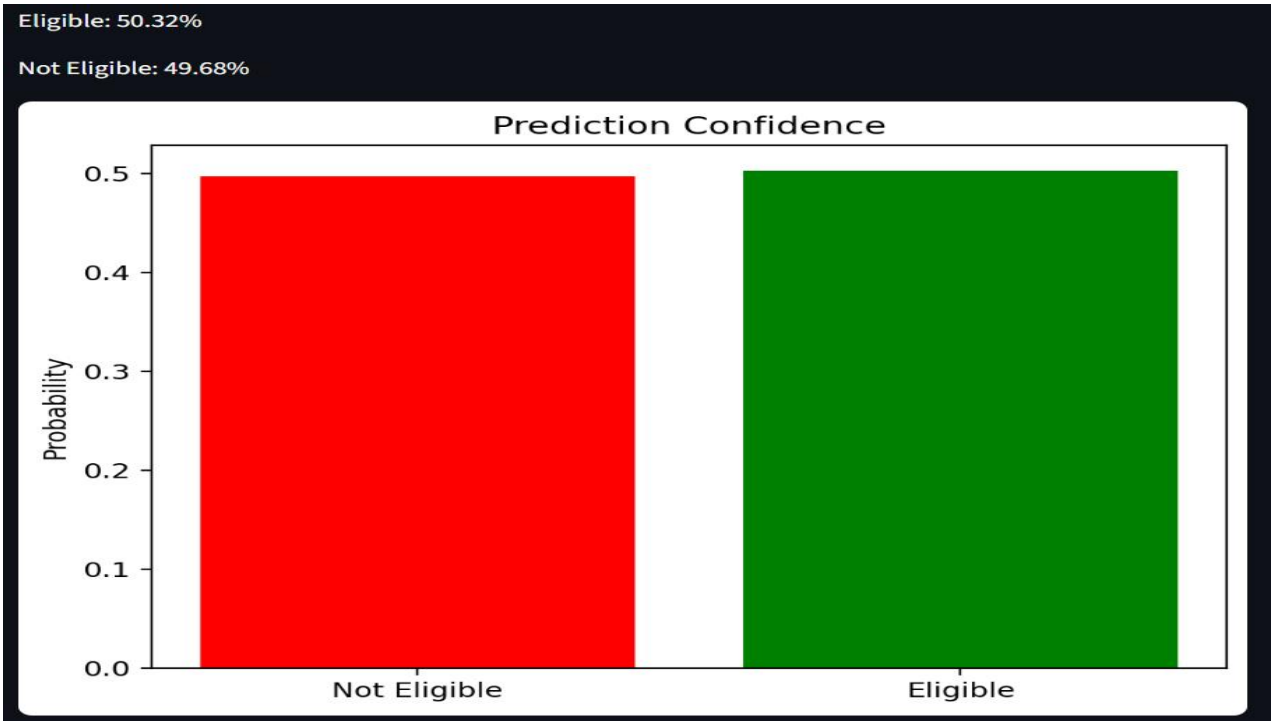
- **Best Performance**: The **Gaussian Naive Bayes** model performs the best with an accuracy of 86.99%, excelling in both precision and recall for class 1 (eligible).
- **Balanced Models**: **KNN** and **Random Forest** also perform strongly with high accuracy and balanced results across classes.
- **Moderate Performance**: The **Logistic Regression** and **SVC** models offer good performance but have slightly lower accuracy and recall for class 1 compared to the other models.

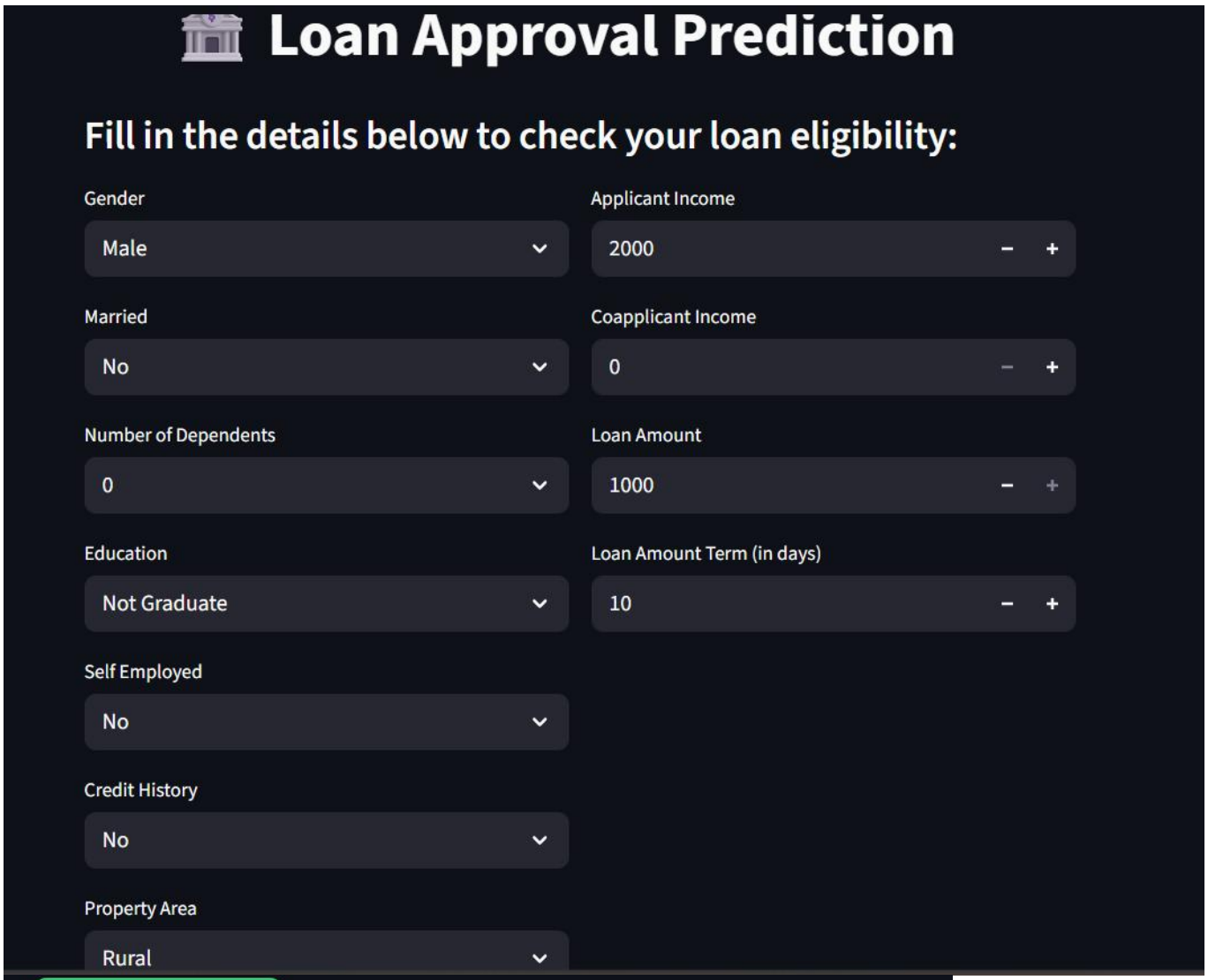
- **Decision Tree**: The **Decision Tree** model has the lowest accuracy (82.14%) and slightly lower precision and recall compared to the other models.

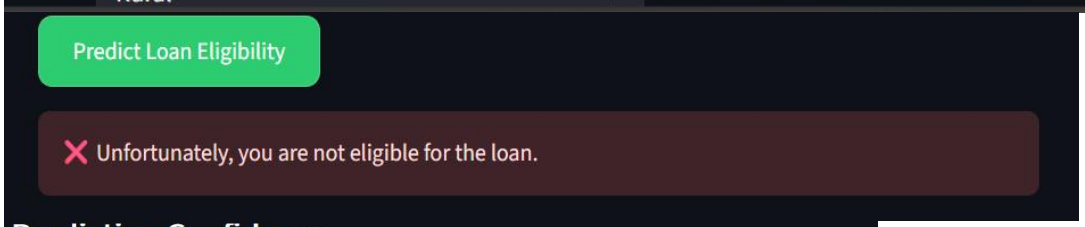
Results and Output:

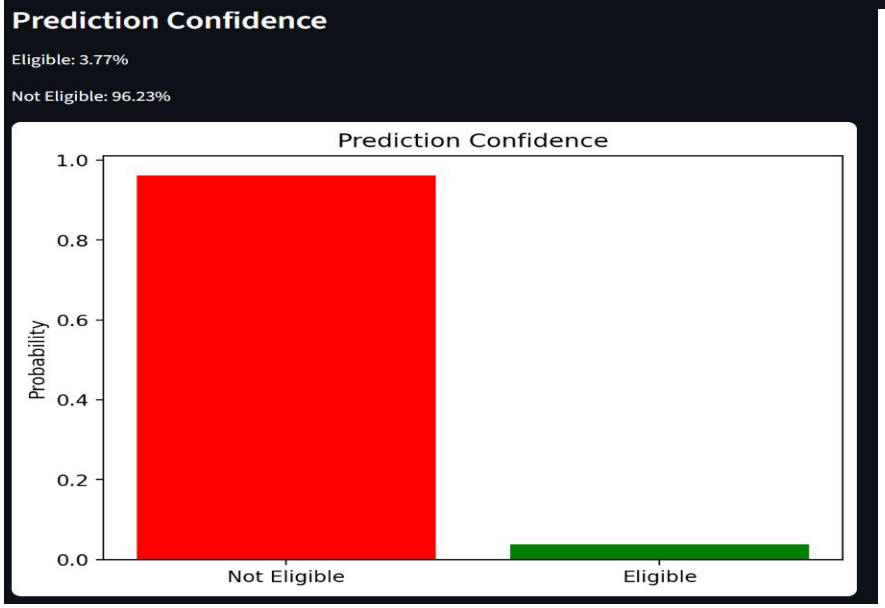







Loan Approval Prediction
Fill in the details below to check your loan eligibility:
Gender
Male
Applicant Income
2000
Married
No
Coapplicant Income
0
Number of Dependents
0
Loan Amount
1000
Education
Not Graduate
Loan Amount Term (in days)
10
Self Employed
No
Credit History
No
Property Area
Rural


Predict Loan Eligibility
Unfortunately, you are not eligible for the loan.


Prediction Confidence
Eligible: 3.77%
Not Eligible: 96.23%
Prediction Confidence
Probability
1.0
0.8
0.6
0.4
0.2
0.0
Not Eligible
Eligible

LOAN APPROVAL PREDICTION

Data Load

```
[ ] #!pip install joblib
```

```
[ ] #!pip uninstall scikit-learn
```

```
[ ] #!pip install scikit-learn==1.2.2
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
#import joblib
warnings.filterwarnings('ignore')
from sklearn.model_selection import KFold,cross_val_score
my_kfold=KFold(n_splits=5)
```

```
[ ] data=pd.read_csv("/content/loan.csv")
    data=data.dropna(how='any',inplace=False)
    print(data)
```

```
       Loan_ID  Gender Married Dependents     Education Self_Employed  \
1     LP001003    Male     Yes          1      Graduate            No
2     LP001005    Male     Yes          0      Graduate           Yes
```

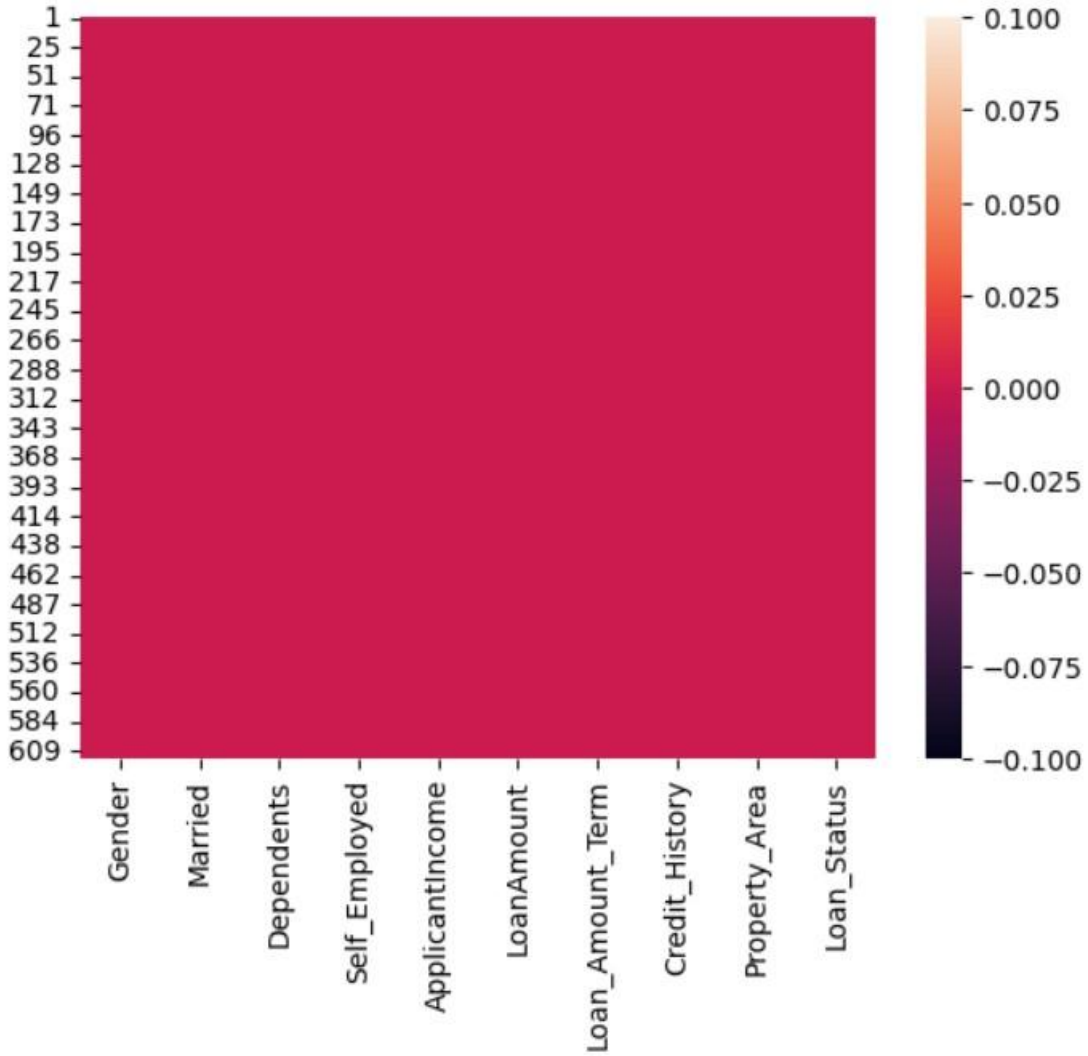
```
..                 ...              ...           ...           ...
609             360.0              1.0         Rural             Y
610             180.0              1.0         Rural             Y
611             360.0              1.0         Urban             Y
612             360.0              1.0         Urban             Y
613             360.0              0.0     Semiurban             N

[480 rows x 10 columns]
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 480 entries, 1 to 613
Data columns (total 10 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Gender            480 non-null    object
 1   Married           480 non-null    object
 2   Dependents        480 non-null    object
 3   Self_Employed     480 non-null    object
 4   ApplicantIncome   480 non-null    int64
 5   LoanAmount        480 non-null    float64
 6   Loan_Amount_Term  480 non-null    float64
 7   Credit_History    480 non-null    float64
 8   Property_Area     480 non-null    object
 9   Loan_Status       480 non-null    object
dtypes: float64(3), int64(1), object(6)
memory usage: 41.2+ KB
```

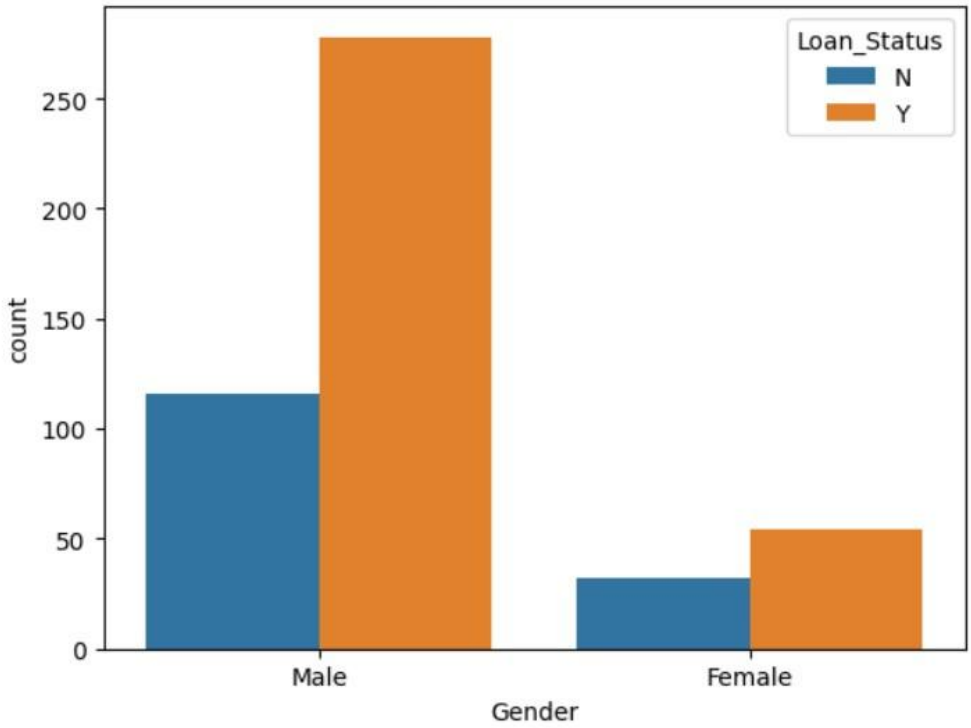
```
sns.heatmap(data.isnull())
```

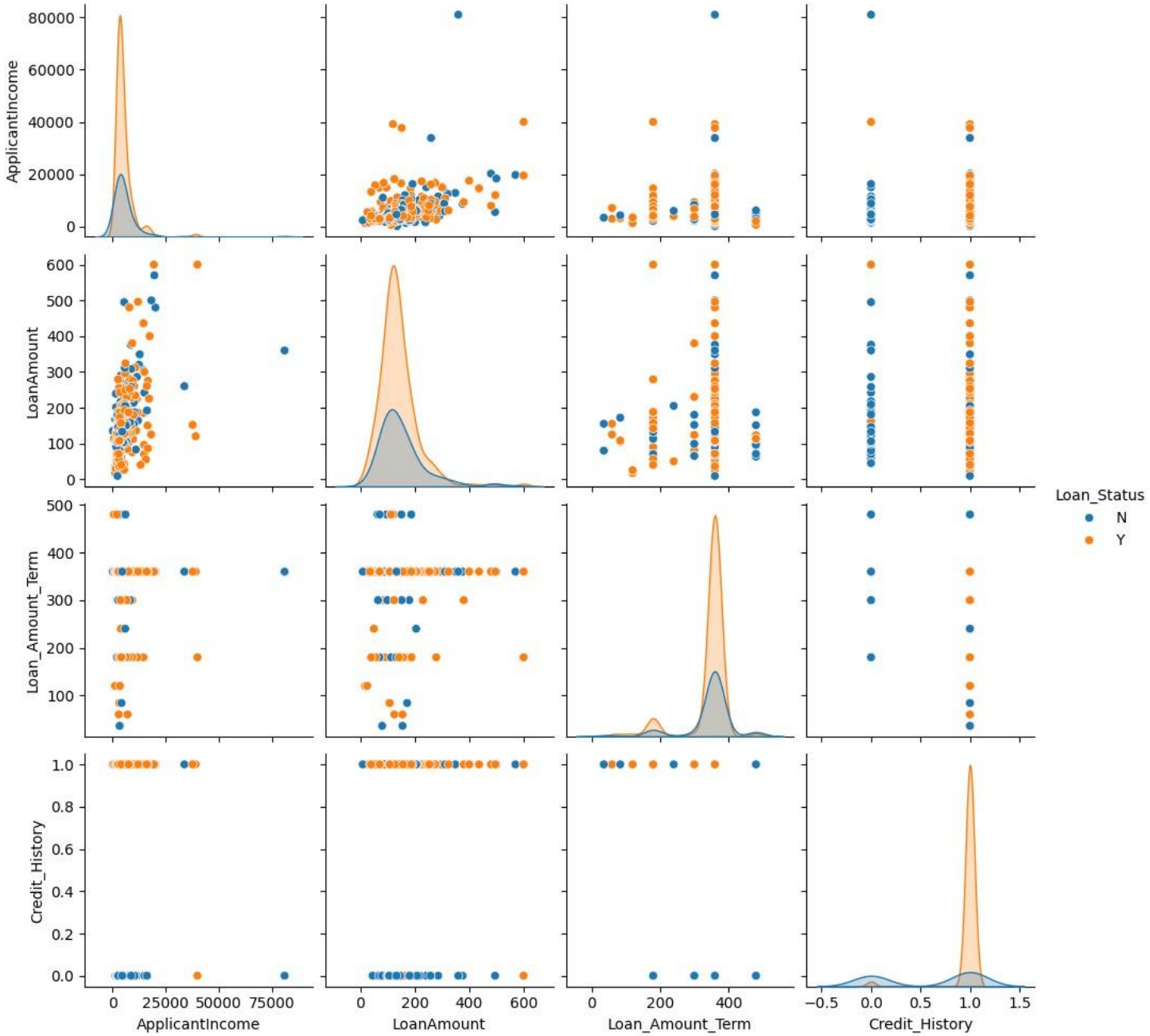
```
<Axes: >
```



```
sns.countplot(x='Gender',data=data,hue='Loan_Status')
```

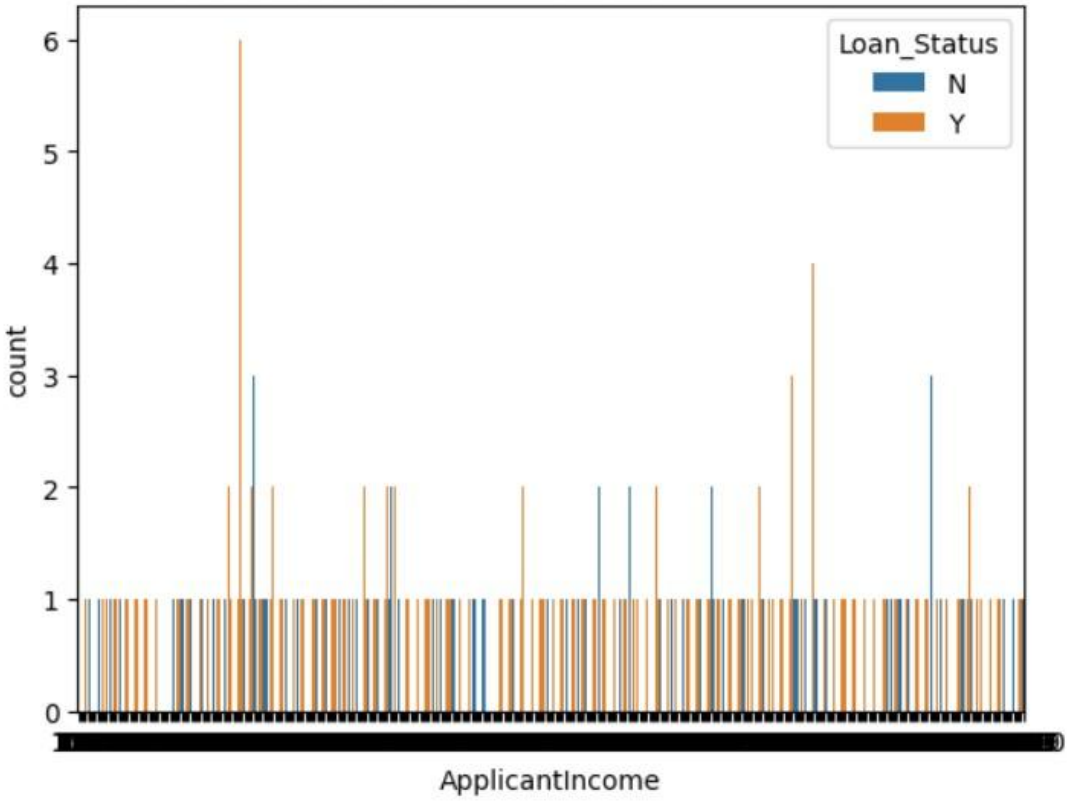
```
<Axes: xlabel='Gender', ylabel='count'>
```




ApplicantIncome
LoanAmount
Loan_Amount_Term
Credit_History
Loan_Status
N
Y

```
sns.countplot(x='ApplicantIncome',data=data,hue='Loan_Status')
```

```
<Axes: xlabel='ApplicantIncome', ylabel='count'>
```



SVM Linear

```
from sklearn.svm import SVC
svm_model=SVC(kernel='linear')
svm_model.fit(x_train,y_train)
```

```
SVC
SVC(kernel='linear')
```

```
y_pred=svm_model.predict(x_test)
ac=accuracy_score(y_test,y_pred)
print(ac)
```

```
0.84375
```

```
#joblib.dump(svm_model,'loan_model.joblib')
```

Random Forest

```
from sklearn.ensemble import RandomForestClassifier
rf=RandomForestClassifier()
rf.fit(x_train,y_train)
```

```
RandomForestClassifier
RandomForestClassifier()
```

```
y_pred=rf.predict(x_test)
ac=accuracy_score(y_test,y_pred)
print(ac)
```

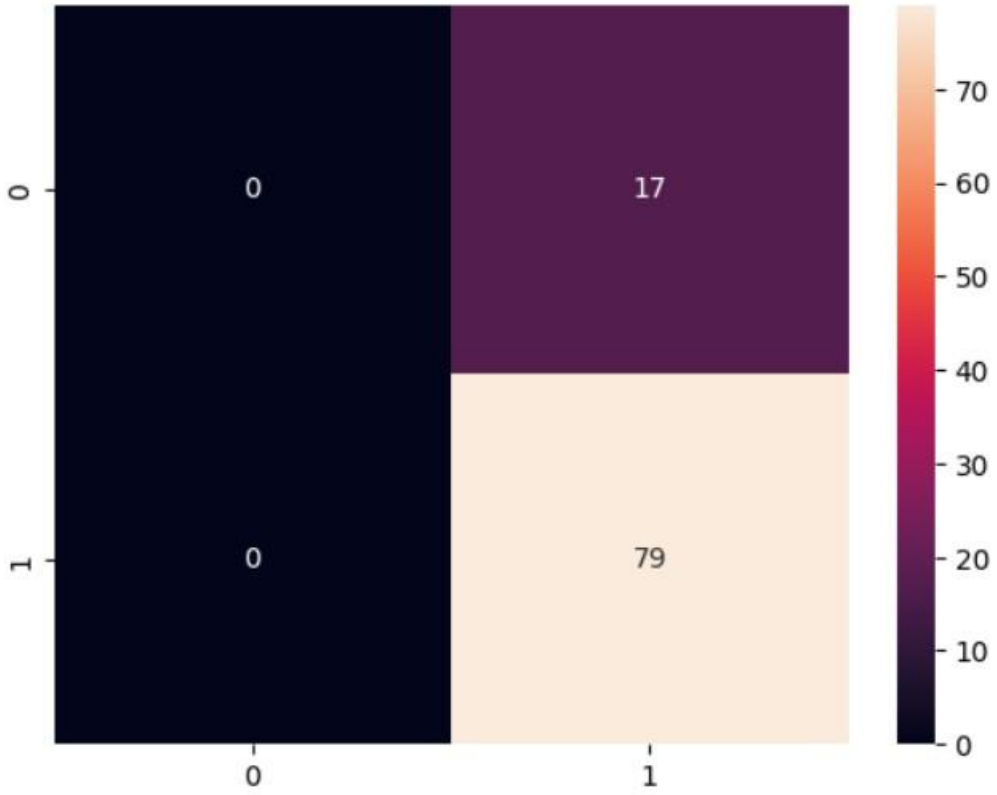
```
0.8125
```

```
myrfsc=cross_val_score(rf,x_test,y_test,cv=my_kfold)
print(myrfsc)
print("Mean of kfold"+str(myrfsc.mean()))
```

```
[0.9        0.89473684 0.78947368 0.73684211 0.78947368]
Mean of kfold0.8221052631578948
```

```
from sklearn.metrics import confusion_matrix,accuracy_score
y_pred=lr.predict(x_test)
cm=confusion_matrix(y_test,y_pred)
print(accuracy_score(y_test,y_pred))
sns.heatmap(cm,annot=True)
```

```
0.8229166666666666
<Axes: >
```



```
[ ] from sklearn.neighbors import KNeighborsClassifier
    knn=KNeighborsClassifier()
    knn.fit(x_train,y_train)
```

```
KNeighborsClassifier
KNeighborsClassifier()
```

```
inp=[[2,3,1,500,200,3,7,1,3]]
yp=knn.predict(inp)
print(yp)
```

```
[1]
```

```
[ ] myknnsc=cross_val_score(knn,x_test,y_test,cv=my_kfold)
    print(myknnsc)
    print("Mean of kfold"+str(myknnsc.mean()))
```

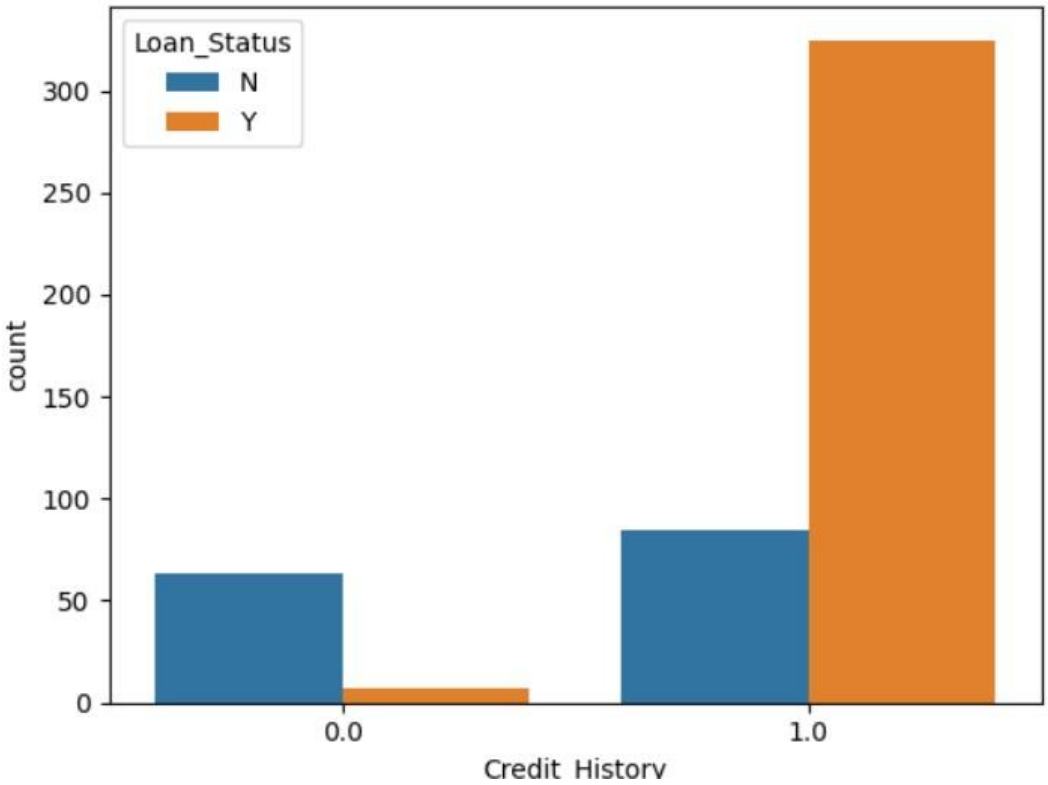
```
[0.9        0.84210526 0.68421053 0.63157895 0.78947368]
Mean of kfold0.7694736842105263
```

```
[ ] from sklearn.linear_model import LogisticRegression
    lr=LogisticRegression()
    lr.fit(x_train,y_train)
```

```
LogisticRegression
LogisticRegression()
```

```
sns.countplot(x='Credit_History',data=data,hue='Loan_Status')
```

```
<Axes: xlabel='Credit_History', ylabel='count'>
```



Chapter 6: Conclusion And Future Scope

CONCLUSION:

The project effectively demonstrates how machine learning can automate and enhance the loan approval process. By using various algorithms on real-world data, the system can accurately predict loan eligibility, reducing manual effort and decision errors. Among all models, Naive Bayes achieved the highest accuracy, showing that even simple algorithms can yield reliable results with proper preprocessing. Machine learning proves to be a valuable tool for financial institutions aiming to make faster and more data-driven loan decisions.

FUTURE SCOPE

The future scope of this project includes several opportunities for extension and improvement. Advanced statistical analysis can be explored by incorporating metrics such as skewness, kurtosis, and hypothesis testing to identify significant factors influencing game lengths. Predictive modeling using machine learning algorithms like regression, decision trees, or neural networks can help predict game outcomes based on key parameters. Real-time analysis capabilities can be introduced with tools like Apache Kafka or Spark, enabling live performance monitoring. Strategy optimization through reinforcement learning or optimization techniques can recommend the best moves or pathways to players. Enhancing data visualization with interactive dashboards using tools like Plotly or Power BI will offer deeper insights and

dynamic analysis. Integrating this project with gaming platforms can automate insights for players and

developers, while scaling the pipeline to handle larger datasets efficiently will ensure broader applicability. Benchmarking game length distributions across various platforms can uncover universal patterns, while automating data collection, cleaning, and visualization will streamline the workflow. Lastly, this methodology can be customized for similar domains such as sports analytics, simulations, or other competitive processes, making the project versatile and impactful.

Chapter 7: References:

References:

- Singh, M., & Jain, S. (2020). *Loan Approval Prediction using Decision Tree Algorithm.* International Journal of Computer Applications, 975–8887.
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- Hossain, M. Z., & Khan, M. R. (2021). *Credit Risk Analysis Using Machine Learning Models*. ACM Digital Library.
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- Patel, P., & Shah, D. (2019). *Loan Approval Prediction Based on Machine Learning Approach.* Proceedings of the IEEE International Conference on Intelligent Systems.